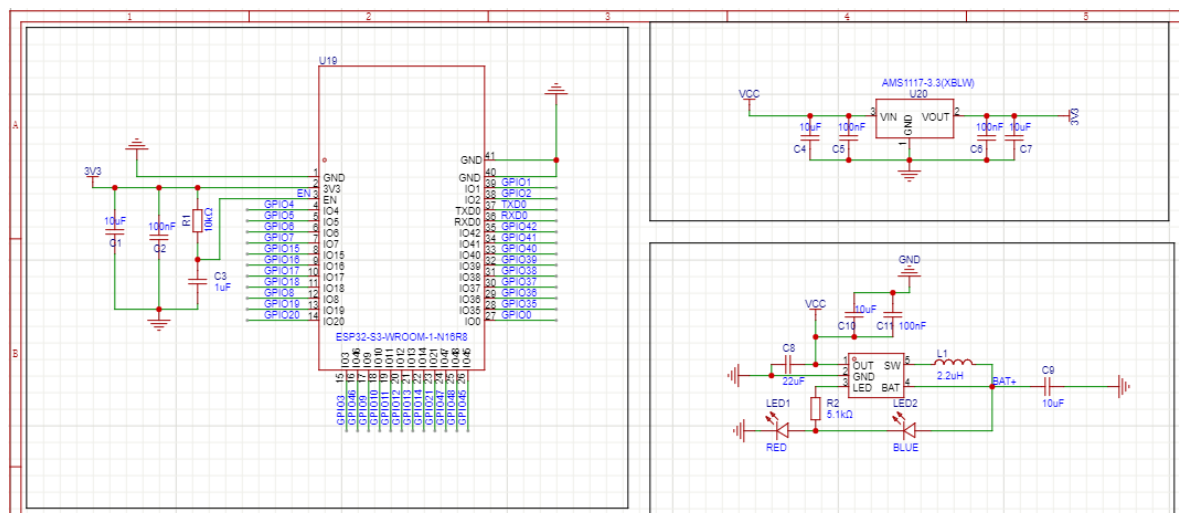
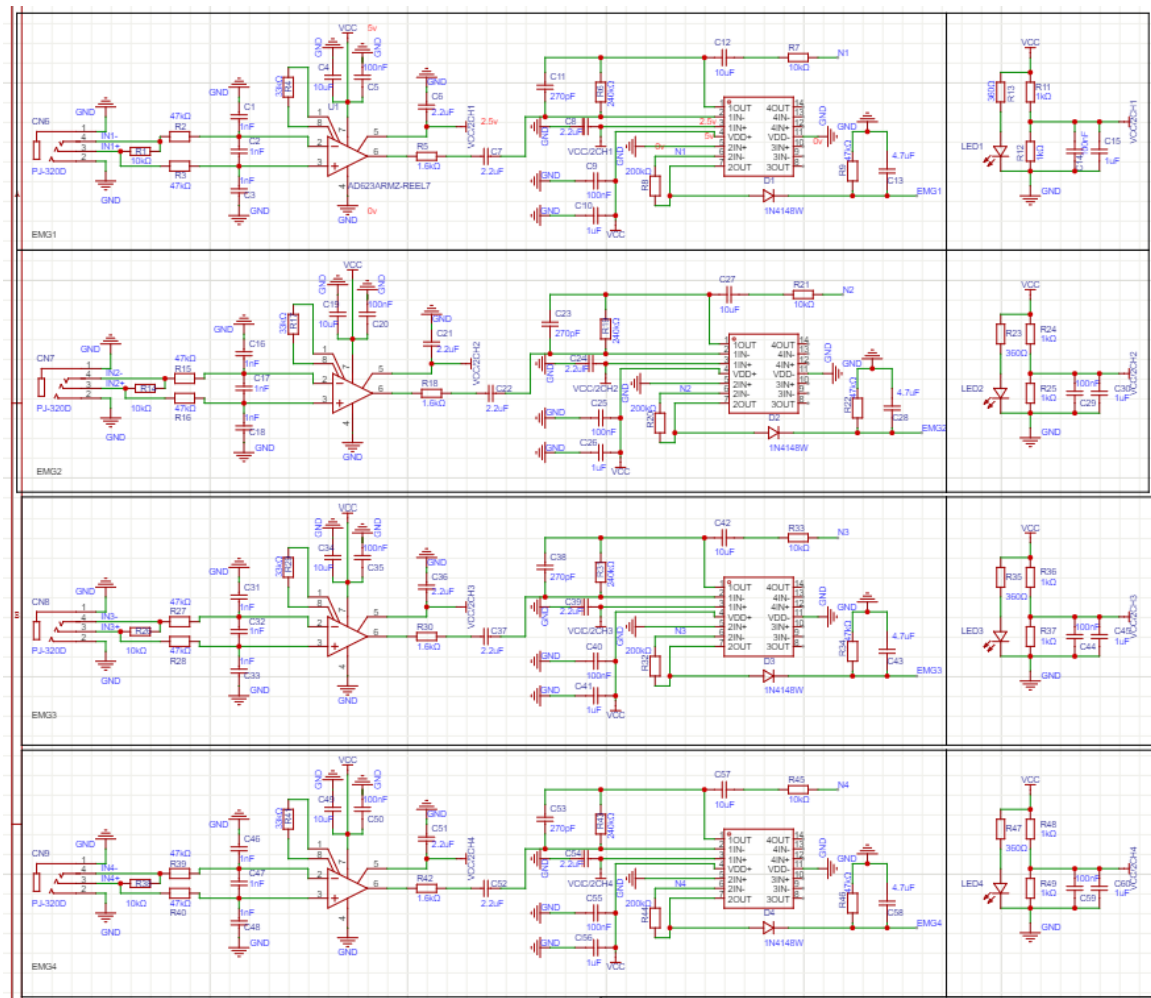


Ballet Dancer Injury Prevention and Technique Correction via Multichannel sEMG and Vision-Guided AI

Incorrect muscle usage is a leading contributor to ballet injuries and poor technique, especially en pointe. This project aims to (1) quantify muscle activation patterns during foundational technique, (2) detect deviation from safe, standard execution in real time, and (3) provide low-burden feedback that helps dancers self-correct before overexertion occurs. The specific objective is to integrate surface electromyography (sEMG) from key muscle groups with a computer vision model that classifies movement quality compared to professionally evaluated exemplars. Currently, ballet lacks accessible tools that provide empirical biomechanical evidence to guide instruction and practice. This project designed and customized a wearable sEMG system embedded in a waistband form factor (see Appendix: circuit and PCB). The device acquires eight sEMG channels targeting the latissimus dorsi, upper trapezius, rectus femoris, and gastrocnemius and streams data to a local web dashboard. A vision pipeline is trained on labeled clips of Royal Academy of Dance barre routines, such as relevé exercises, to distinguish between correct and incorrect executions as defined by instructors. The study analyzes sEMG signal amplitudes to identify muscle-use patterns associated with common faults, including, but not limited to, sickling, hip hikes, and delayed calf activation. A functional prototype reliably records multichannel sEMG during practice and aligns muscle activation patterns with video for post-hoc review. Early pilot sessions with four senior-level ballet dancers indicate that the system can capture distinct activation signatures between correct and

incorrect trials and that dancers tolerate the waistband without noticeable interference during barre work. If validated with a larger cohort, this approach could enable studio-floor screening and objective progress tracking, helping dancers and instructors make data-informed adjustments before pain or injury emerges.

Appendix A: Circuit Schematic Design



Appendix B: PCB

